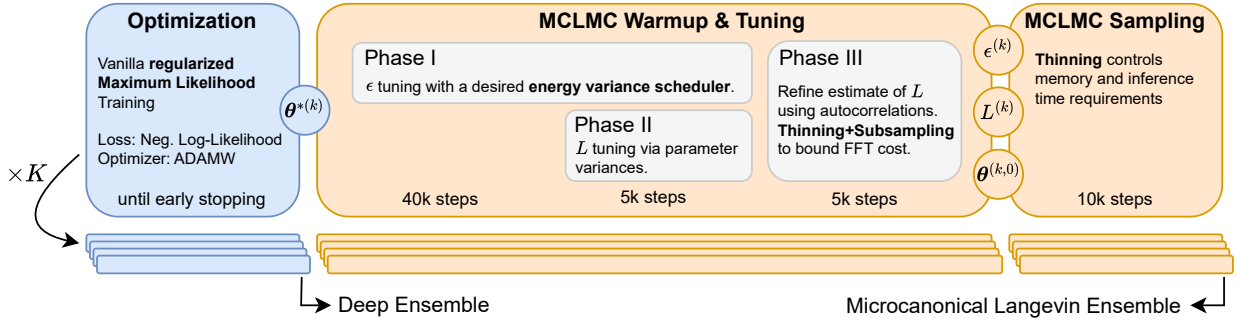
Optimization
Vanilla **regularized Maximum Likelihood** Training
Loss: Neg. Log-Likelihood
Optimizer: ADAMW
until early stopping
$\times K$
$\boldsymbol{\theta}^{*(k)}$
MCLMC Warmup & Tuning
Phase I
ϵ tuning with a desired **energy variance scheduler**.
40k steps
Phase II
L tuning via parameter variances.
5k steps
Phase III
Refine estimate of L using autocorrelations. **Thinning+Subsampling** to bound FFT cost.
5k steps
$\epsilon^{(k)}$
$L^{(k)}$
$\boldsymbol{\theta}^{(k,0)}$
MCLMC Sampling
Thinning controls memory and inference time requirements
10k steps
Deep Ensemble
Microcanonical Langevin Ensemble